MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Website:- mathsmanthan.co.in

Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 12 (SOLUTIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, all intermediate algebraic steps, and clearly boxed final answers.
2. In internal choice (**OR**) questions, both options are solved completely with identical step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M, 3M]**.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. The 30th term of the AP $10, 7, 4, \ldots$ is: [1]

(A) 97 (B) 77 (C) -77 (D) -87

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(C) -77** [0.5 Mark]

Given: $a = 10$, $d = 7 - 10 = -3$, $n = 30$.

Formula / Theorem Used: $a_n = a + (n-1)d$

Justification: [0.5 Mark]

$a_{30} = 10 + (30-1)(-3) = 10 - 87 = \mathbf{-77}$.

Q2. The middle term of the finite AP $213, 205, 197, \ldots, 37$ is: [1]

(A) 125 (B) 117 (C) 121 (D) 129

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) 125** [0.5 Mark]

Given: $a = 213$, $d = -8$, $l = 37$.

Formula / Theorem Used: Middle term of odd AP $= \dfrac{a+l}{2}$

Justification: [0.5 Mark]

$37 = 213 - 8(n-1) \implies 8(n-1) = 176 \implies n = 23$ (odd); $a_{12} = 213 + 11(-8) = \frac{213+37}{2} = \mathbf{125}$.

Q3. The sum of all three-digit numbers divisible by 7 is: [1]

(A) 70336 (B) 70350 (C) 70314 (D) 70320

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) 70336** [0.5 Mark]

Given: AP is $105, 112, \ldots, 994$ ($a = 105$, $d = 7$, $l = 994$).

Formula / Theorem Used: $n = \dfrac{l-a}{d} + 1$ and $S_n = \dfrac{n}{2}(a+l)$

Justification: [0.5 Mark]

$n = \frac{994-105}{7} + 1 = \frac{889}{7} + 1 = 128$; $S_{128} = 64(105 + 994) = 64 \times 1099 = \mathbf{70336}$.

Q4. The sum of the first 20 odd natural numbers is: [1]

(A) 100 (B) 200 (C) 400 (D) 420

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: (C) 400 [0.5 Mark]

Using the formula for the first n odd natural numbers ($a = 1$, $d = 2$):

Formula / Theorem Used: $S_n = n^2$

Justification: [0.5 Mark]

$S_{20} = \frac{20}{2}[2(1) + 19(2)] = 10(2 + 38) = 10 \times 40 = 20^2 = \mathbf{400}$.

Q5. **Assertion (A):** In an AP with last term l and common difference d, the nth term from the end is $l - (n-1)d$.
Reason (R): Counting backwards from l reverses the AP so its common difference becomes $-d$. [1]

(A) Both (A) and (R) are true, and (R) is correct explanation.

(B) Both (A) and (R) are true, and (R) is not correct explanation.

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: (A) Both (A) and (R) are true, and (R) explains (A). [0.5 Mark]

Reversing a finite AP:

Formula / Theorem Used: Reversed AP: $l,\ l-d,\ l-2d,\ \ldots$ gives n^{th} term $= l + (n-1)(-d) = l - (n-1)d$

Justification: [0.5 Mark]

Reason (R) identifies the negative common difference when read backward (true), proving Assertion (A).

MATHS MANTHAN • TUITION CLASSES •

SECTION B: Very Short Answer Questions — MATHS MANTHAN | (2 × 2 = 4 Marks)

Q6. If the sum of the first n terms of an AP is $S_n = 2n^2 + 3n$, find its 4th term. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: $S_n = 2n^2 + 3n$.

Formula / Theorem Used: $a_4 = S_4 - S_3$

Step 1: Evaluating S_4 and S_3 [1 Mark]

$S_4 = 2(4)^2 + 3(4) = 32 + 12 = 44; \quad S_3 = 2(3)^2 + 3(3) = 18 + 9 = 27.$

Step 2: Finding a_4 [1 Mark]

$$a_4 = S_4 - S_3 = 44 - 27 = \boxed{17}$$

Q7. Find the sum of the first 20 terms of an AP in which $d = 7$ and $a_{20} = 143$. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Finding the First Term a [1 Mark]

$a_{20} = a + 19d = 143 \implies a + 19(7) = 143 \implies a + 133 = 143 \implies a = 10.$

Formula / Theorem Used: $S_n = \frac{n}{2}(a + a_n)$

Step 2: Computing S_{20} [1 Mark]

$$S_{20} = \frac{20}{2}(10 + 143) = 10 \times 153 = \boxed{\mathbf{1{,}530}}$$

SECTION C: Short Answer Questions — MATHS MANTHAN | (2 × 3 = 6 Marks)

Q8. In an AP, the 8th term is three times its 2nd term and the 12th term exceeds twice its 5th term by 1. Find the AP. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: $a_8 = 3a_2$ and $a_{12} = 2a_5 + 1$.

Formula / Theorem Used: $a_n = a + (n-1)d$

Step 1: Translating Conditions into Linear Equations [1.5 Marks]

$a + 7d = 3(a + d) \implies a + 7d = 3a + 3d \implies 4d = 2a \implies a = 2d.$ — (i)

$a + 11d = 2(a + 4d) + 1 \implies a + 11d = 2a + 8d + 1 \implies 3d = a + 1.$ — (ii)

Step 2: Solving for d, a, and Forming the AP [1.5 Marks]

Substituting $a = 2d$ into (ii): $3d = 2d + 1 \implies d = 1$, hence $a = 2(1) = 2$.

Required AP: **2**, **3**, **4**, **5**, ...

Q9. Find the sum of all natural numbers between 250 and 1000 that are divisible by 3. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Identifying the AP and Finding n [1.5 Marks]

Multiples of 3 between 250 and 1000: $252, 255, 258, \ldots, 999$ ($a = 252$, $d = 3$, $l = 999$).

Formula / Theorem Used: $a_n = a + (n-1)d$ and $S_n = \frac{n}{2}(a + l)$

$$252 + (n-1)(3) = 999 \implies 3(n-1) = 747 \implies n - 1 = 249 \implies n = 250$$

Step 2: Computing S_{250} [1.5 Marks]

$$S_{250} = \frac{250}{2}(252 + 999) = 125 \times 1{,}251 = \boxed{\mathbf{1{,}56{,}375}}$$

Q10. In an AP, if S_4, S_8, S_{12} denote the sums of its first 4, 8, and 12 terms respectively, prove that $S_{12} = 3(S_8 - S_4)$. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Given: Let first term be a and common difference be d.

Formula / Theorem Used: $S_n = \frac{n}{2}[2a + (n-1)d]$

Step 1: Writing Expressions for S_4, S_8, and S_{12} [2 Marks]

$S_4 = \frac{4}{2}(2a + 3d) = 2(2a + 3d) = 4a + 6d.$

$S_8 = \frac{8}{2}(2a + 7d) = 4(2a + 7d) = 8a + 28d.$

$S_{12} = \frac{12}{2}(2a + 11d) = 6(2a + 11d) = 12a + 66d.$

Step 2: Simplifying RHS $= 3(S_8 - S_4)$ [3 Marks]

$$3(S_8 - S_4) = 3[(8a + 28d) - (4a + 6d)] = 3(4a + 22d) = 12a + 66d = \boxed{\mathbf{S_{12}} \quad \text{[Hence Proved]}}$$

Q11. If the sum of the first 7 terms of an AP is 49 and that of its first 17 terms is 289, find the sum of its first n terms (S_n) and hence find S_{25}. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Given: $S_7 = 49$ and $S_{17} = 289$.

Formula / Theorem Used: $S_n = \frac{n}{2}[2a + (n-1)d]$

Step 1: Forming Two Linear Equations in a and d [1.5 Marks]

$S_7 = \frac{7}{2}(2a + 6d) = 49 \implies 7(a + 3d) = 49 \implies a + 3d = 7.$ — (i)

$S_{17} = \frac{17}{2}(2a + 16d) = 289 \implies 17(a + 8d) = 289 \implies a + 8d = 17.$ — (ii)

Step 2: Solving for d and a [1.5 Marks]

Subtracting (i) from (ii): $5d = 17 - 7 = 10 \implies d = 2$. Substituting $d = 2$ into (i): $a + 6 = 7 \implies a = 1$.

Step 3: Finding S_n and S_{25} [2 Marks]

$$S_n = \frac{n}{2}[2(1) + (n-1)(2)] = \frac{n}{2}(2 + 2n - 2) = \frac{n}{2}(2n) = n^2$$

Setting $n = 25$: $S_{25} = 25^2 = 625$.

$$\boxed{\mathbf{S_n = n^2}, \qquad \mathbf{S_{25} = 625}}$$

Q12. CASE STUDY: Decorative Stepping Stones in a Landscaped Walkway

A garden designer lays a path of 40 engraved stepping stones sequentially numbered in an arithmetic progression: $7, 11, 15, 19, \ldots$ [5]

(i) Find the number engraved on the 25th stepping stone. [1]

(ii) Which stepping stone bears the number 115? [1]

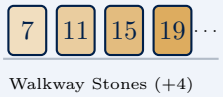
(iii) Compute the total sum of all numbers engraved across the 40 stepping stones. [3]

— **OR** —

If only 31 stones are used instead, what number is engraved on the exact central stone? [3]

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Extracted Scenario Data: Stone engravings form the AP $7, 11, 15, 19, \ldots$ with $a = 7$, $d = 11 - 7 = 4$, and $n = 40$.



Walkway Stones (+4)

Formula / Theorem Used: $a_n = a + (n-1)d$ and $S_n = \frac{n}{2}[2a + (n-1)d]$

Sub-part (i): Number on the 25th Stepping Stone [1 Mark]

$a_{25} = 7 + (25-1)(4) = 7 + 24 \times 4 = 7 + 96 = \boxed{\mathbf{103}}$.

Sub-part (ii): Position of Stone Engraved with 115 [1 Mark]

$7 + (n-1)(4) = 115 \implies 4(n-1) = 108 \implies n - 1 = 27 \implies n = \boxed{\mathbf{28}^{\text{th}} \text{ stone}}$.

Sub-part (iii): Sum of All 40 Engraved Numbers [3 Marks]

$$S_{40} = \frac{40}{2}[2(7) + (40-1)(4)] = 20(14 + 39 \times 4) = 20(14 + 156) = 20 \times 170 = \boxed{\mathbf{3{,}400}}$$

Sub-part (iii) [OR Option]: Central Stone When 31 Stones Are Laid [3 Marks]

For $n = 31$ (odd), the central stone is the $\left(\frac{31+1}{2}\right)^{\text{th}} = 16^{\text{th}}$ stone.

$$a_{16} = 7 + (16-1)(4) = 7 + 15 \times 4 = 7 + 60 = \boxed{\mathbf{67}}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★